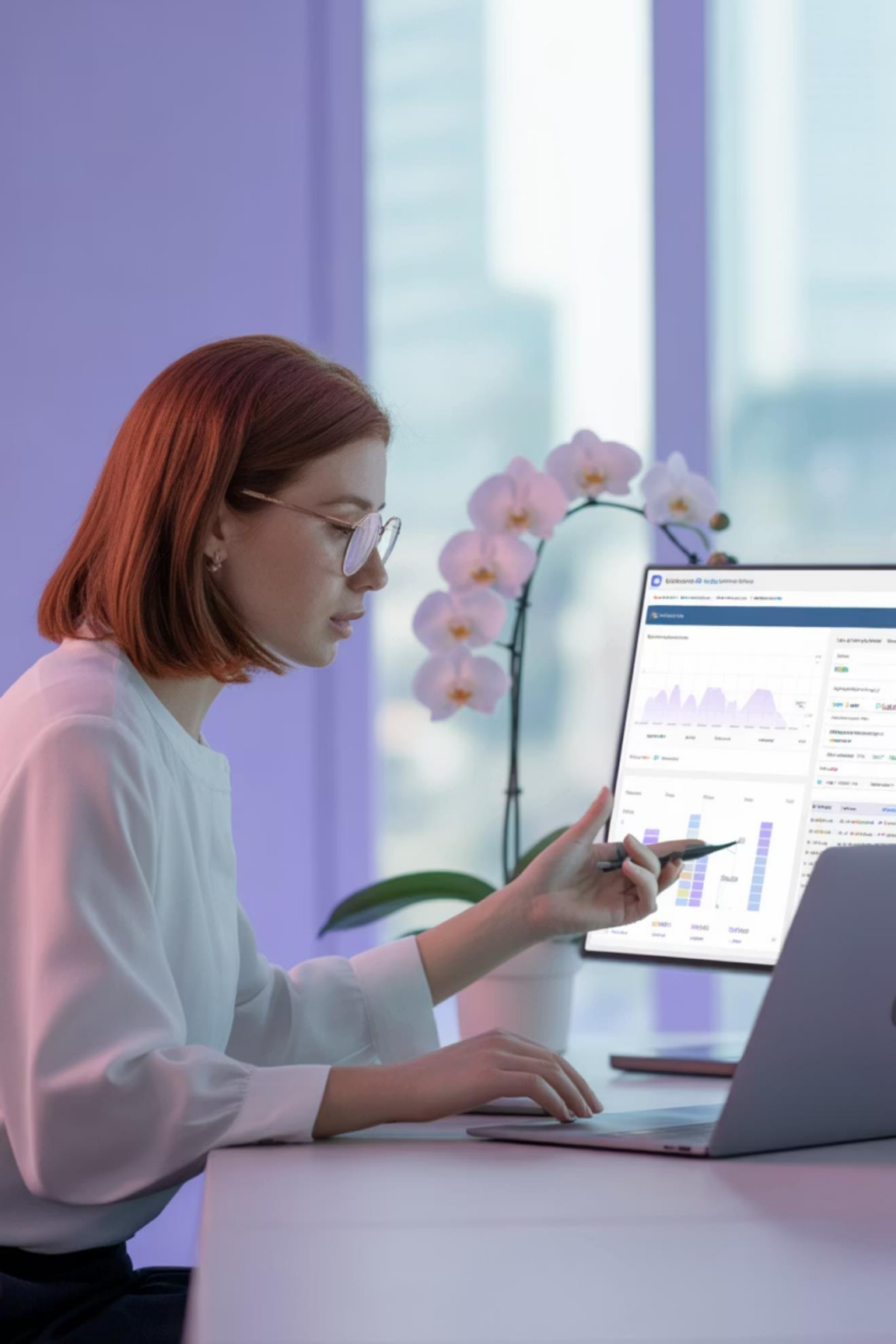




Recall Topics Covered in Previous lecture

- Familiarizing with Excel
- Revisit data/variables
- Descriptive statistical analyses with real-world data
 - Numerical analysis
 - Visualization
 - Tabulation

Understanding Relevance

- Overview of different types of variables and their functions in research.
- Different types of level of measurement
- Two branches and methods of statistics
- How statistical methods are chosen

What is Variables in Research?

- In research, a **variable** is any characteristic, trait, or condition that can change or take on different values.

Example:

- **Gender:** can varies among Male (M) and Female (F). In modern society, the value be more.
- **Height:** can be 170.5 cm, 170.55 cm, etc.
- **Weight:** can be measured in kilograms with or without decimal values.
- **Test Scores:** Often whole numbers, like 85, 90, or 95, depending on the scoring system.



Types of Variables: Overview

To capture the **Essence** of variables, we need to view them from different **Perspectives** or **Angles**: nature, value, role in research, ...



Independent and Dependent Variables

The terms commonly used in experimental study but have been becoming popular in correlational study for simplicity.

Independent Variables

- Manipulated or categorized by the researcher
- Causes or influences change in other variables

Example: Marketing campaign or expenditure

Dependent Variables

- Measured in response to independent variable
- The outcome or effect being studied

Example: Purchase intention or brand awareness



Example: In a marketing study, you might manipulate the independent variable (ad design) to measure its effect on the dependent variable (consumer purchase intent).

Predictors vs. Outcomes

Predictor Variables

- Used to forecast or explain outcome variables
- Similar to independent variables but broader context
- May include variables not manipulated by researcher

Outcome Variables

- The response or result being measured
- Similar to dependent variables
- The "what happens" in your research question

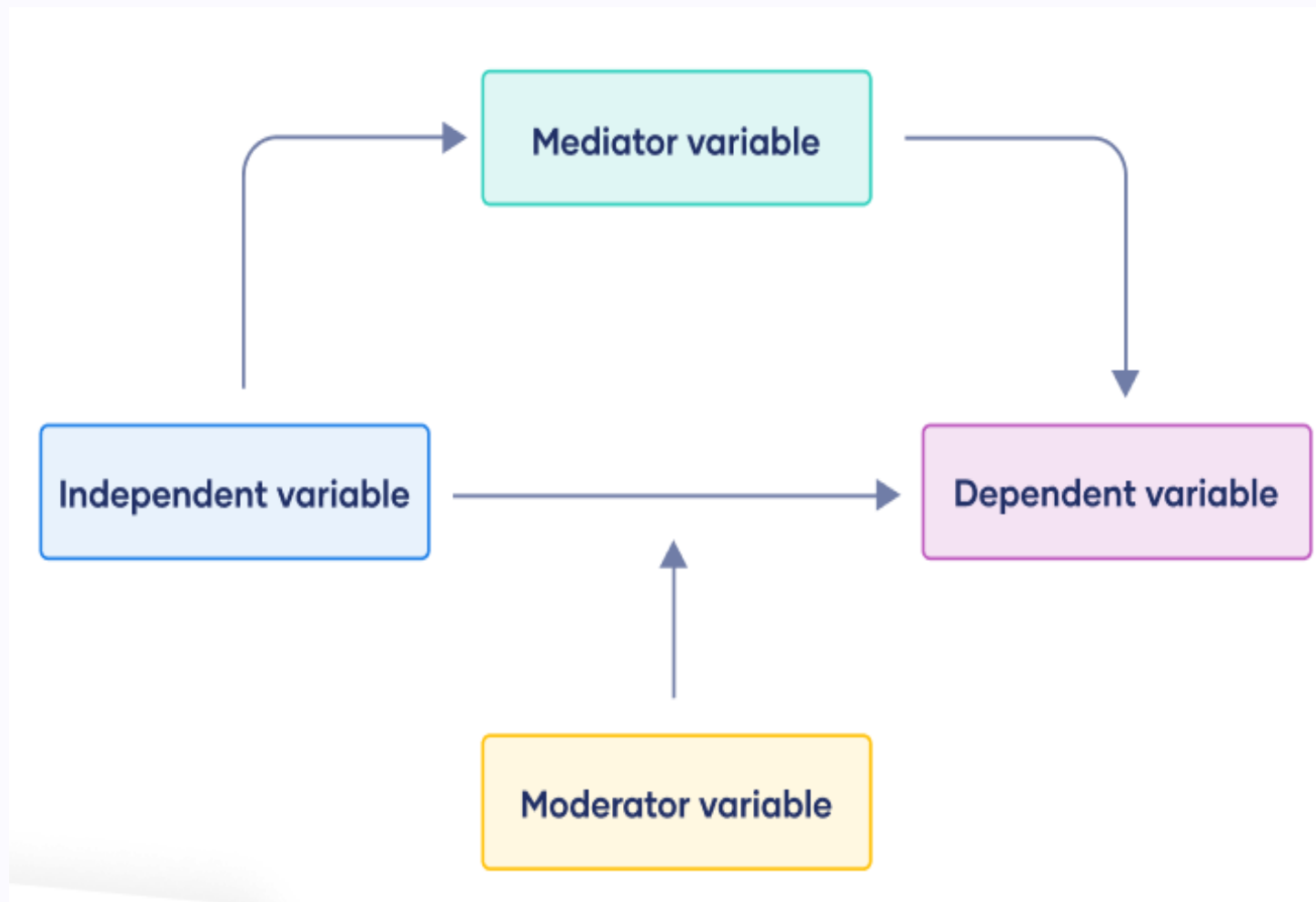


Example: In a study to examine the customer satisfaction:

- **Predictors:** might include service quality, product functionality or living location ...
- **Outcome:** is variable measuring the overall satisfaction scores on a specific product.

Mediating & Moderating Variables

These two variables, mediating and moderating, help explain how or why an independent variable influences a dependent variable. Mediators clarify the causal pathway, while moderators indicate the conditions under which this relationship is strengthened or weakened, providing deeper insights.



Example: A study aims to examine how advertising (independent variable) affects sales (dependent variable). In this case:

- *Brand Awareness* serves as a mediating variable that explains the process through which advertising influences sales.
- *Customer Engagement* acts as a moderating variable that affects the strength of this relationship.

Numerical Variable

A **numerical variable** (or quantitative variable) is a type of variable that represents measurable quantities. These variables can take on a wide range of values and are typically expressed in numerical form. Numerical variables are divided into two main types:

1. Continuous Variables

A variable that can take any value within a given range and can be measured with great precision.

- **Interval:** Equal intervals but no true zero (e.g., temperature in °C)
- **Ratio:** Equal intervals with true zero (e.g., weight, time, distance)

2. Discrete Variables

A variable that can only take specific, distinct values and are often counts of items or occurrences.

- **Number of Children:** You can have 0, 1, 2, etc., but not 2.5 children.
- **Number of Cars:** Counts like 1, 2, or 3 cars, but not fractions.
- **Scores on a Test:** Often whole numbers, like 85, 90, or 95, depending on the scoring system.

Categorical Variables

Categorical variables represent distinct types, groups, or categories without inherent numerical meaning. They are qualitative in nature.



Nominal Variables

- Categories with no inherent order or ranking

Examples: Product types, brand names, ethnicity, blood type, religion etc.



Ordinal Variables

- Categories with a meaningful ranked order

Examples: Customer satisfaction (low to high), educational level, social class



Binary/Dichotomous

- Special case with exactly two categories

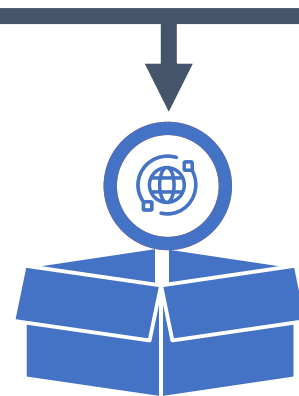
Examples: Responses such as Yes/No, True/False, Purchase/No, Purchase, Male/Female

Types of Levels of Measurement



Level of Measurement

A classification that describes the nature of information within the values assigned to the variables.

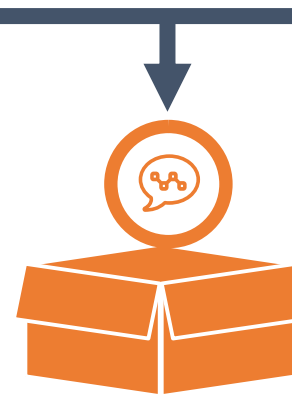


Nominal

Groups, categories ... without order.

Example:

- ⇒ Gender
- ⇒ Nationality
- ⇒ Ethnicity
- ⇒ Language
- ⇒ Etc.

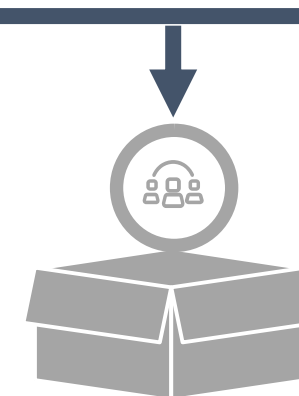


Ordinal

Groups, categories but with certain order.

Example:

- ⇒ Education level
- ⇒ Likert Scales
- ⇒ Rank/Position ...



Interval

Number with no *True Zero*.

Example:

- ⇒ Temperature
- ⇒ SAT Score, GMAT ...
- ⇒ Etc.



Ratio

Number with essence of True Zero.

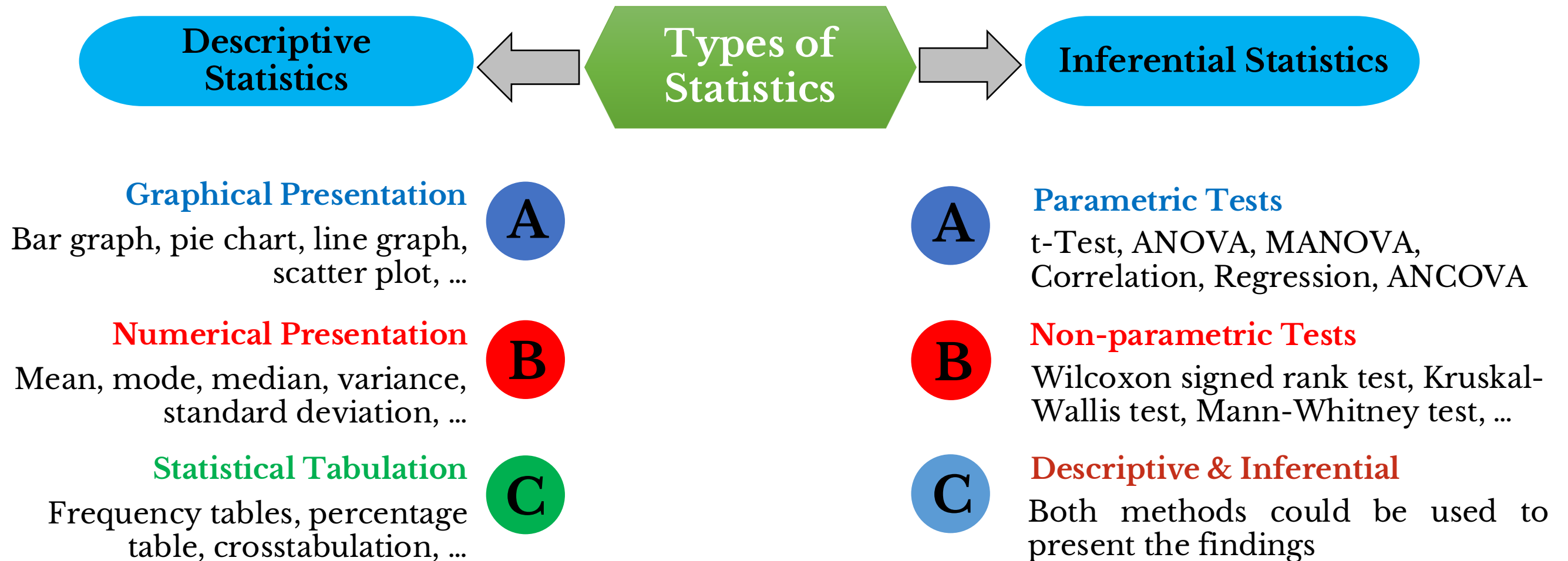
Example:

- ⇒ Age, height
- ⇒ Length, distance
- ⇒ Income, revenue
- ⇒ Etc.

Note: In SPSS, we have only three measurement data: Nominal, Ordinal and Scale (Interval & Ratio)

Types of Statistical Methods

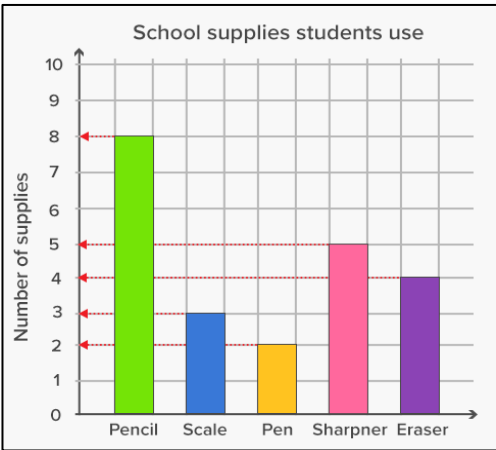
Statistics is a range of procedures for gathering, organizing, analyzing, presenting and interpreting quantitative data.



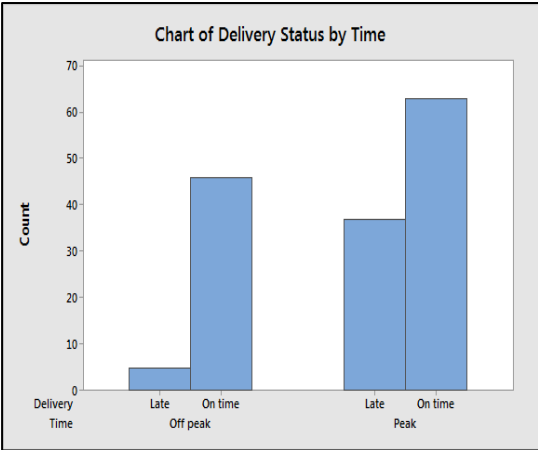
Common Descriptive Statistical Methods

Variable/Data	Purpose	Chart/Graph	Statistical Summary
One Categorical Nominal & Interval)	Shows frequencies/ proportions/percentages	Pie-Bar Chart	Class/Type percentages
Two Categorical	Compares proportions within groups	Stacked/Multiple Bar	Percentages within groups
One Scale	Shows distribution of results	Histogram	Mean and Standard deviation
Two Scale	Shows relationship between two variables and helps detect outliers	Scatter Plot	Correlation coefficient
One Scale/One Categorical	Compares spread of values	Boxplot	Median and IQR
Scale by time	Displays changes over time Comparison of groups	Line Chart	Means by time point

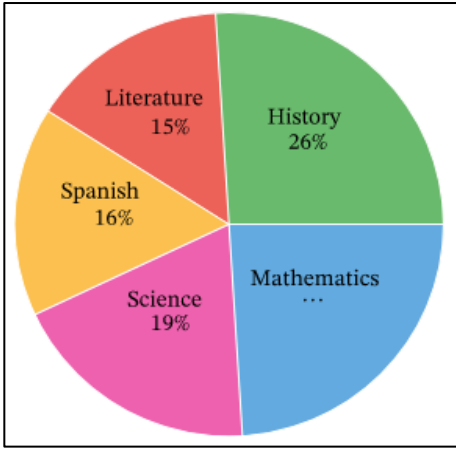
SNAPSHOTS: Descriptive Statistics Presentation



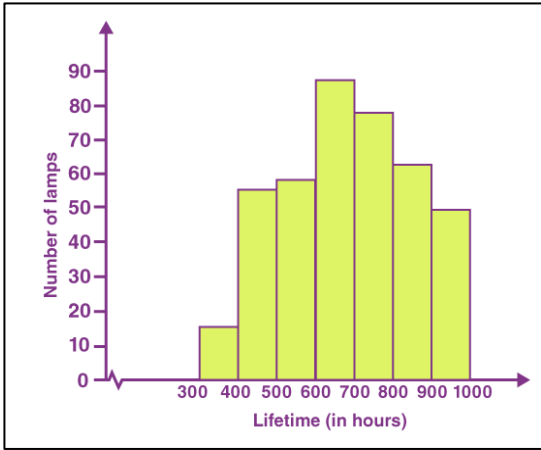
Bar Chart



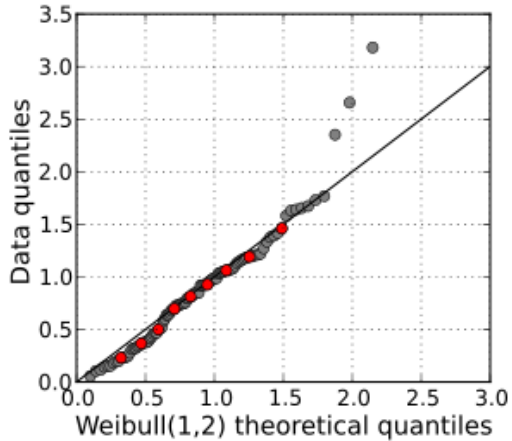
Grouped Bar Chart



Pie Chart



Histogram



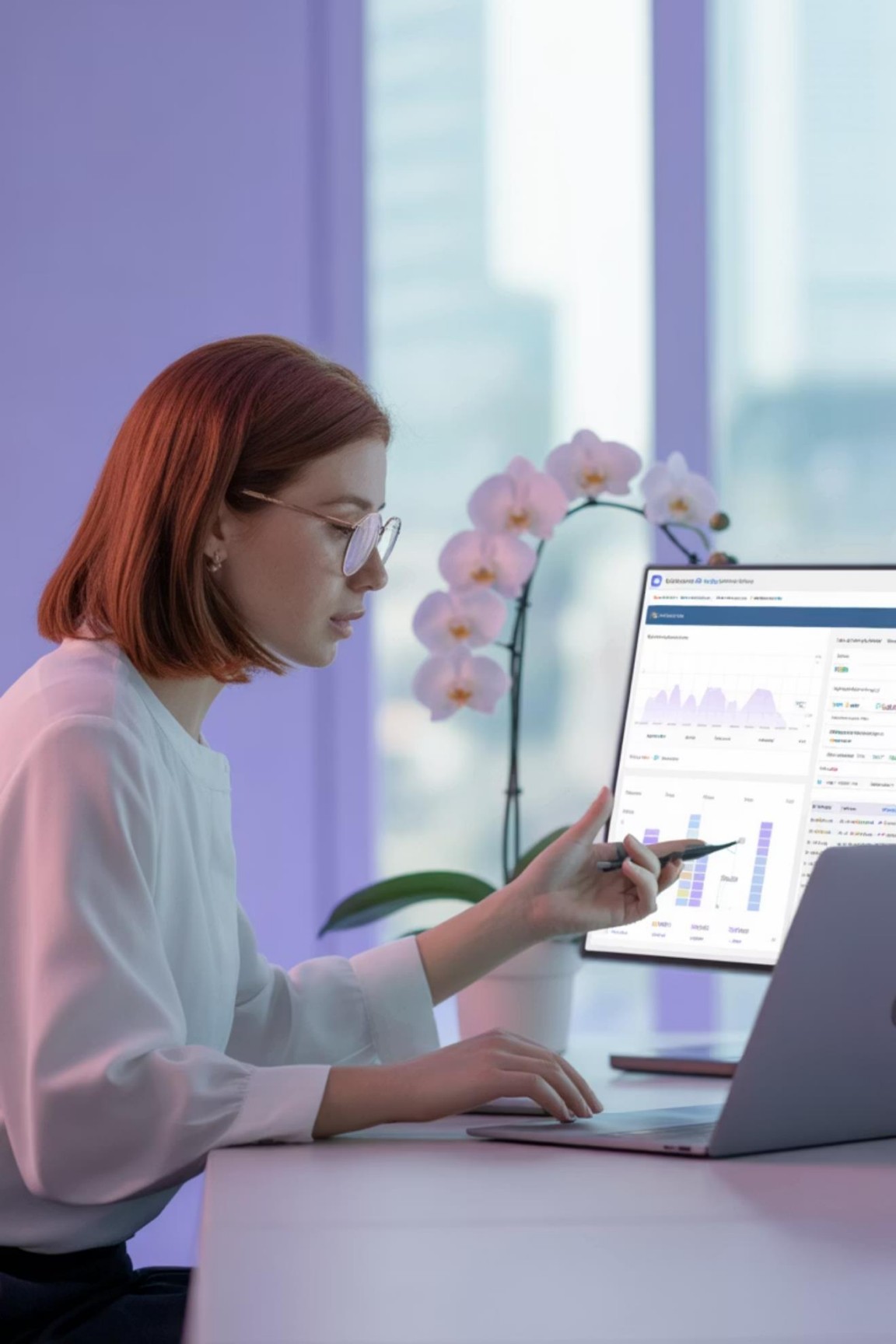
QQ-Plot

Table 4: Distribution of participants based on education level and their profession

	Under-graduate	Post-graduate	Total
Self Employed	37	12	49
Public Servant	15	57	72
Private/NGO Staff	48	43	91
Total	100	112	212

Table 3: Respondents' overseas experience

	Frequency	Percent
Had Overseas Experience	172	81.1
No Overseas Experience	40	18.9
Total	212	100.0



Hypotheses Testing

Topics in This Session:

- Familiarizing with Excel
- Revisit data/variables
- Descriptive statistical analyses with real-world data
 - Numerical analysis
 - Visualization
 - Tabulation

Common Statistical Tests and Their Applications

Understanding Terms Used:

- Qualitative data, categorical data, nominal data (here coded as **QUAL**) and they refer to the same thing as following:
 - Nominal or Ordinal: group with or without order [**e.g. Gender (M/F), Education Level (No Educ/Primary/Secondary/Higher Education)**]
 - **Binary**: group without order and can be only two or more values [**e.g. Gender (M/F), Agreement (Y/N)**].
- Quantitative data, scale data, continuous data (here coded as **QUAN**) and they refer to the same thing.

Variable		Statistical Test	Objectives	Example
IV	DV			
Known value	QUAN	One-Sample t-Test	Compares the mean of a single sample to a known value or a population mean.	Testing if the average height of a sample of students is different from the national average height. (IV: known value, DV: Height)
Binary	QUAN	Independent t-Test	Compares the means of two independent groups to see if they are significantly different.	Comparing test scores of students from two different schools. (IV: School, DV: Test Score)

Common Statistical Tests and Their Applications

Variable		Statistical Test	Objectives	Example
IV	DV			
2 Times/ 2 Conditions	QUAN	Paired Sample t-Test	Compares means from the same group at different times or under different conditions.	Testing the effectiveness of a new teaching method by comparing students' scores before and after the method is applied. [IV: Condition (Pre/Post Test), DV: Test Score]
QUAL (3+ Values)	QUAN	One-Way ANOVA	Compares means of three or more independent groups.	Examining the effect of three different diets on weight loss in participants. [IV: Diet (Diet 1/Diet 2/Diet 3), DV: Weight]
3 Times/ 3 Conditions	QUAN	Repeated Measures ANOVA	Compares means when the same subjects are used for each group (repeated measures).	Analyzing the effect of different doses of a medication on blood pressure in the same group of patients. [IV: Dose (Dose 1/Dose 2/Dose 3), DV: Health Condition]

Common Statistical Tests and Their Applications

Variable		Statistical Test	Objectives	Example
IV	DV			
QUAL	QUAL	Chi-Square Test	Tests the association between two categorical variables.	Determining if there is a relationship between gender and voting preference in an election. [IV: Gender (Male/Female), DV: Voting (Party A/Party B/Party C)]
QUAN	QUAN	Correlation	Measures the strength and direction of the linear relationship between two scale variables.	Investigating the relationship between hours studied and exam scores among students. [IV: Hour Study (Number), DV: Test Score (Number)]
QUAN	QUAN	Simple Linear Regression	Models the relationship between two quantitative variables.	Predicting a student's final grade based on the number of hours studied. [IV: Hour Study (Number), DV: Test Score (Number)]

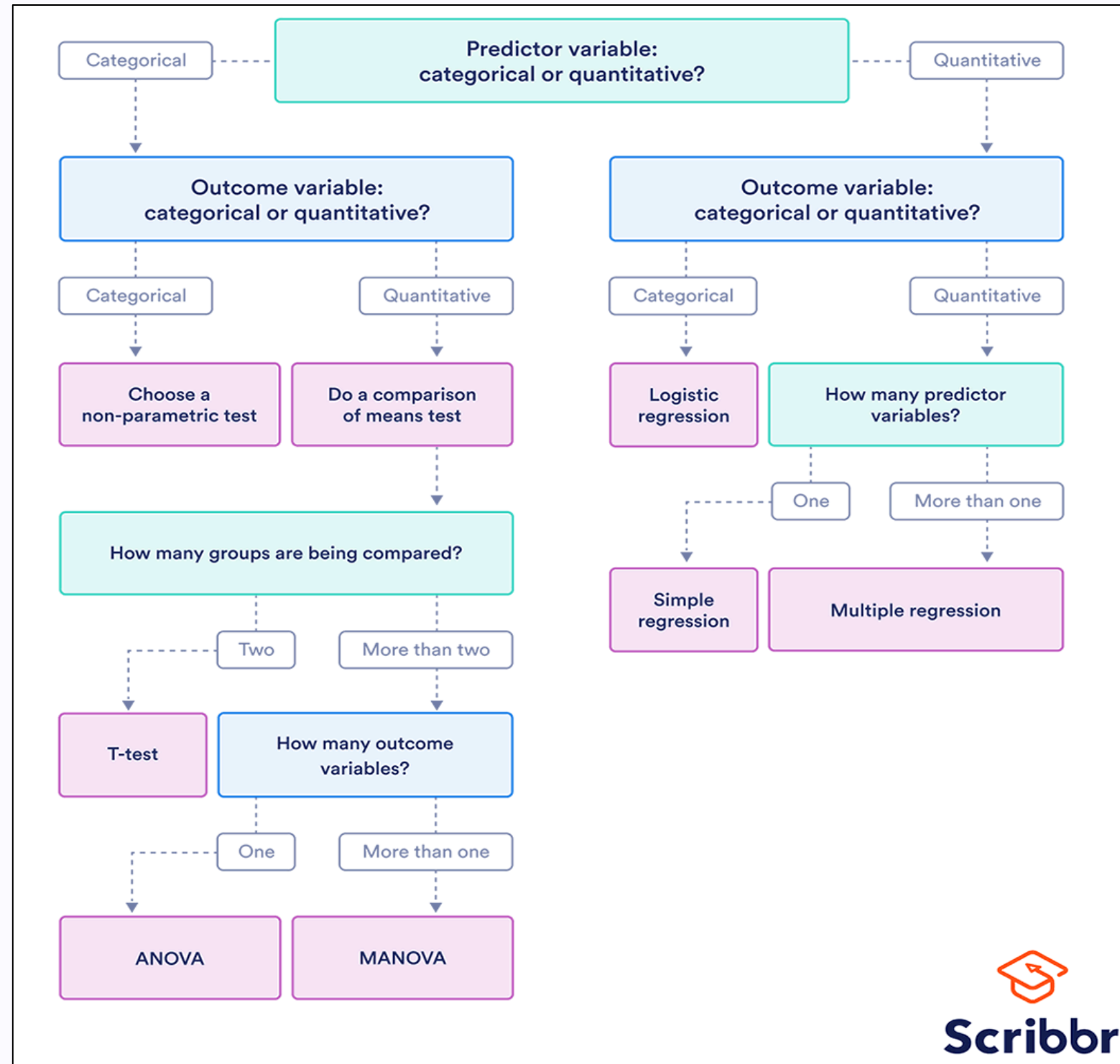
Common Statistical Tests and Their Applications

Variable		Statistical Test	Objectives	Example
IV	DV			
ANY	QUAL	Multiple Regression	Models the relationship between a dependent variable and two or more independent variables.	Examining how factors (study hours, attendance, and homework completion) affect students' grades. [IVs: Study Hour, Attendance, Homework, DV: Test Score]
ANY	Binary	Logistic Regression	Model used to predict the probability of a binary outcome based on independent variables.	Predict customer churn by analyzing purchase history, demographics ... to identify at-risk customers. [IVs: Total Purchases, Total Revenue, Average Order Value, Gender; DV: Customer Churn (Churned = 1, Not Churned = 0)]
Replacement of Independent t-Test when data not <i>“Normal”</i>		Mann-Whitney U-Test	A non-parametric test for comparing two independent groups.	Comparing test scores of students from two different schools. (IV: School, DV: Test Score)

Common Statistical Tests and Their Applications

Variable		Statistical Test	Objectives	Example
IV	DV			
Replacement of Paired sample t-Test when data not “ <i>Normal</i> ”		Wilcoxon Signed-Rank Test	Non-parametric test for comparing two related samples.	Testing the effectiveness of a new teaching method by comparing students' scores before and after the method is applied. [IV: Condition (Pre/Post Test), DV: Test Score]
Replacement of One-Way ANOVA when data not “ <i>Normal</i> ”		Kruskal-Wallis Test	Non-parametric alternative to one-way ANOVA for comparing three or more groups.	Examining the effect of three different diets on weight loss in participants. [IV: Diet (Diet 1/Diet 2/Diet 3), DV: Weight]

Flowchart for Choosing Statistical Tests



Business Scenario: Overview

A company has been tracking customer satisfaction scores for its range of products over the past few years. As the management team prepares for strategic planning, they want to evaluate whether customer satisfaction has remained consistent this year compared to previous years. The satisfaction scores are collected through surveys, and to make informed decisions about product enhancements and marketing strategies, the team aims to analyze the current year's satisfaction scores against the historical data.

Guiding Direction/Questions:

- Using exploratory data analysis (descriptive statistical methods), to assess the general characteristics of the dataset?
 - What is the distribution of customer satisfaction scores for this year? Also, the characteristics of other variables.
 - What is the relational tendency of customer satisfaction with regard to **Income, Purchase Amount, Brand Loyalty** etc.?

Understanding Scenario & Dataset

Practice Dataset: **Customer's Satisfaction**

No.	Variable	Measurement	No.	Variable	Measurement
1	ID	Scale	14	Media_Influence	Ordinal/Nominal
2	Age	Scale	15	Discount_Sensitivity	
3	Gender	Nominal	16	Return_Rate	
4	Income	Ordinal	17	Customer_Satisfaction	Scale
5	Marital_Status	Nominal	18	Engagement_withAds	
6	Education	Ordinal	19	Device_UsedShop	Nominal
7	Purchase_Category	Nominal	20	Payment_Method	Nominal
8	Purchase_Amount	Scale	21	Discount_Used	Nominal
9	Frequent_Purchase	Scale	22	Customer_Loyalty	Scale
10	Purchase_Channel	Nominal	23	Purchase_Intent	Nominal
11	Brand_Loyalty	Scale	24	Shipping_Preference	Nominal
12	Product_Rating	Scale	25	Time_Decision	Scale
13	Time_onProduct	Scale			

Defining Statistical Methods

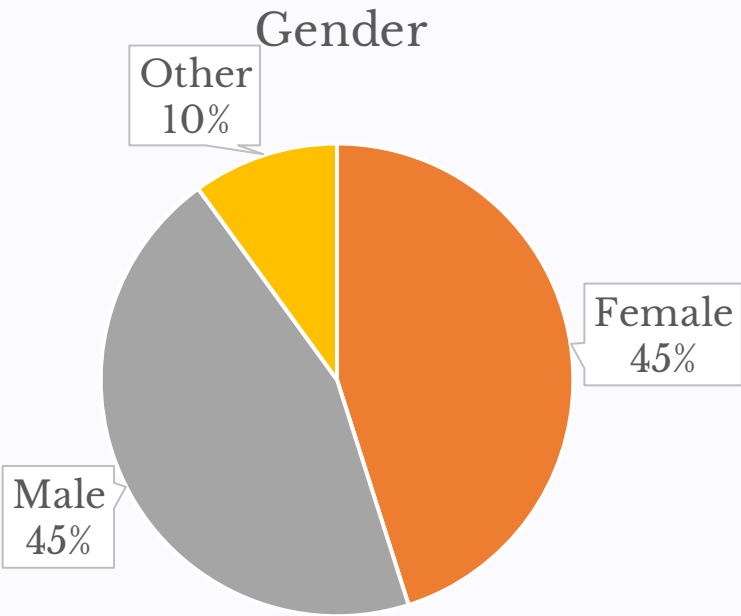
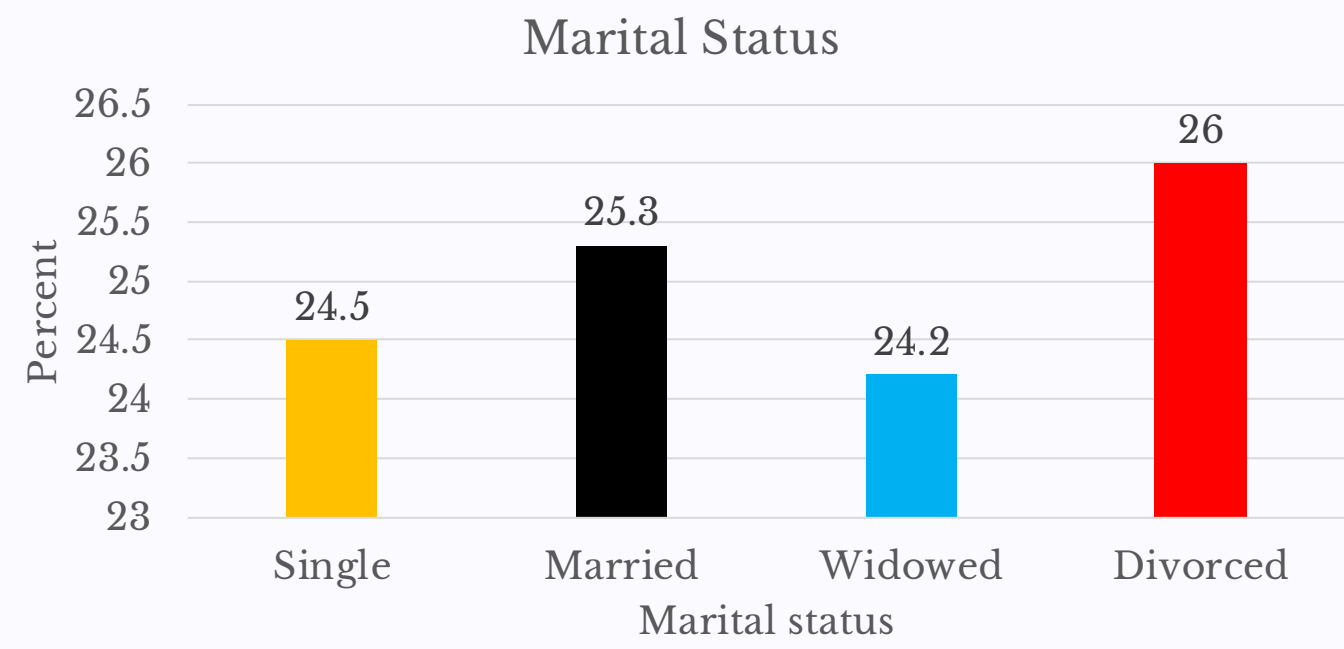
Variable Groups	
Quantitative Variables	Qualitative Variables
Age, Purchase_Amount, Frequent_Purchase, Brand_Loyalty, Product_Rating, Time_onProduct, Customer_Satisfaction, Time_Decision	Gender, Income, Marital_Status, Education, Purchase_Category, Purchase_Channel, Media_Influence, Discount_Sensitivit, Return_Rate, Engagement_withAds, Device_UsedShop, Payment_Method, Discount_Used, Customer_Loyalty, Purchase_Intent, Shipping_Preference

Analysis Methods				
	Univariate Analysis		Bivariate Analysis	
	QUAN	QUAL	Variable Group	Method
Numerical Methods	Mean, Mode, Median, SD, Var, ...	-	QUAN vs QUAN	Scatterplot,
Visualization Method	Histogram, Density	Bar, Pie, ...	QUAN vs QUAL	Boxplot,
Tabulation Method	Descriptive Table	Frequency Table	QUAL vs QUAL	Clustered Bar Chart, Stacked Bar Chart,

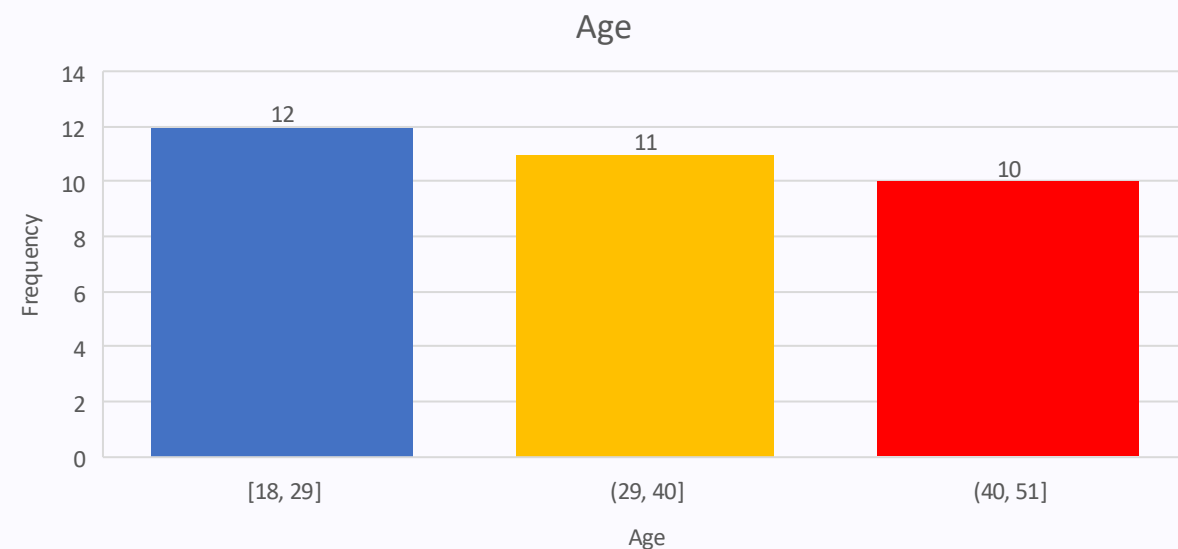
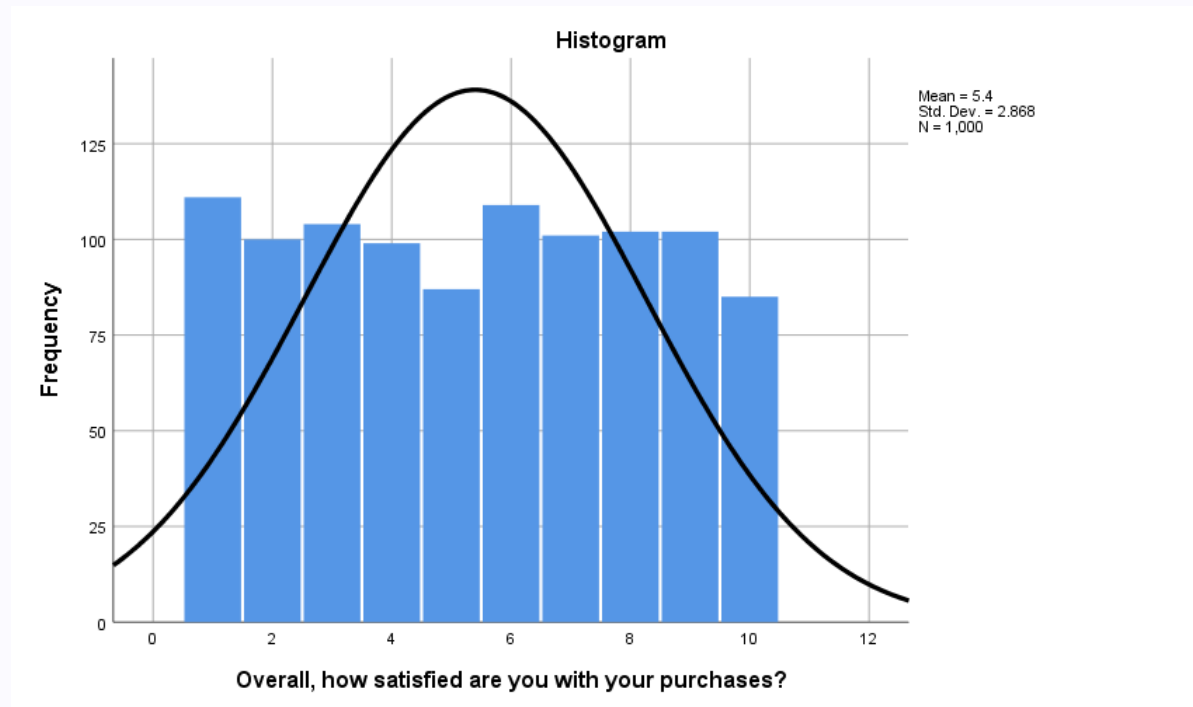
Selected Practices Results

Gender					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Female	451	45.1	45.1	45.1
	Male	449	44.9	44.9	90.0
	Other	100	10.0	10.0	100.0
	Total	1000	100.0	100.0	

Marital status					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Single	245	24.5	24.5	24.5
	Married	253	25.3	25.3	49.8
	Widowed	242	24.2	24.2	74.0
	Divorced	260	26.0	26.0	100.0
	Total	1000	100.0	100.0	



Interpretation & Recommendations



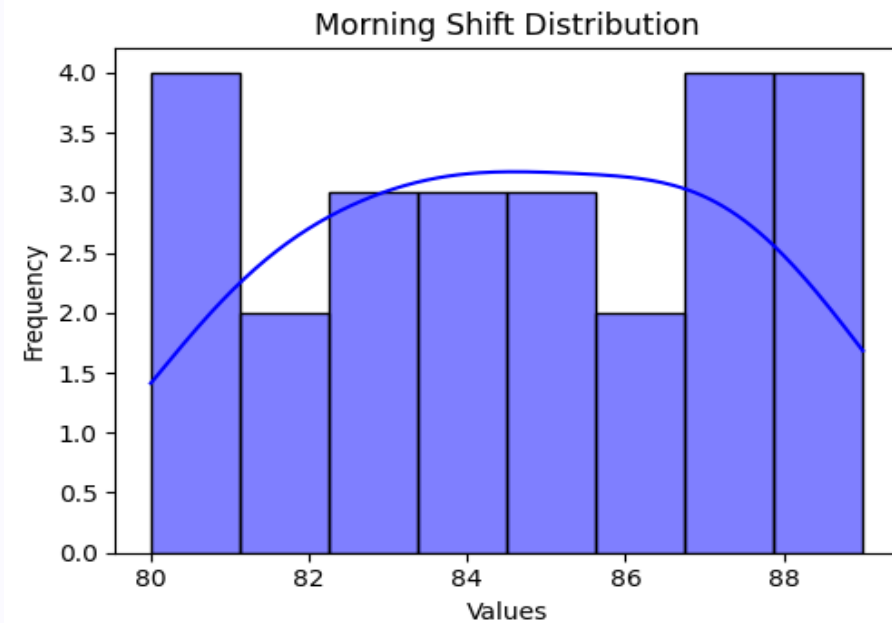
Statistics				
		Time to Purchase	Age	Time Researching
N	Valid	1000	1000	1000
	Missing	0	0	0
Mean		250.79	34.30	1.0130
Median		223.35	34.50	1.0000
Mode		106	27	1.00
SD		137.092	9.353	.79180
Skewness		.517	-.003	-.002
SE of Skewness		.077	.077	.077
Kurtosis		-.586	-1.175	-1.429
SE of Kurtosis		.155	.155	.155
Minimum		9	18	.00
Maximum		851	50	2.00

Questions to Consider

- Does knowing the values of central tendency enough to make an inform decision? If **YES**, why? If **NO**, why?
 - If **NO**, what are other information we should know more? Provide example why these additional information are needed?
- Consider the following scenario:

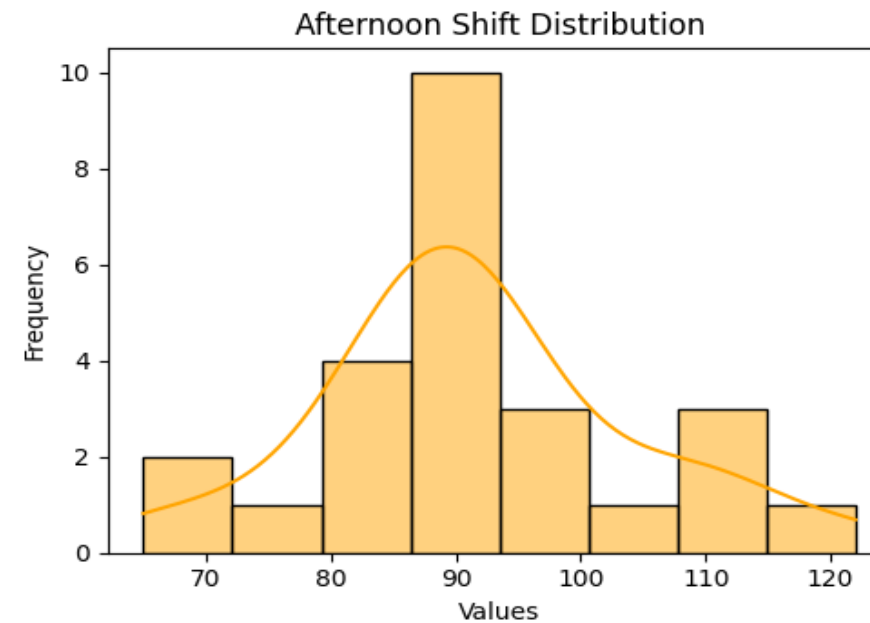
Group	Data (Productivity for the last 20 days)	Measures
Morning Shift	80, 81, 81, 82, 84, 84, 85, 87, 83, 85, 83, 87, 88, 86, 84, 89, 86, 83, 81, 82, 87, 85, 89, 88, 87	Mean: 84.68, Median: 85.0, Mode: 87
Afternoon	65, 96, 95, 92, 78, 110, 71, 83, 104, 85, 86, 87, 88, 89, 90, 90, 122, 93, 110, 96, 82, 90, 110, 88, 90	Mean: 91.6, Median: 90.0, Mode: 90
Night	120, 120, 99, 110, 72, 85, 87, 120, 72, 74, 76, 78, 80, 85, 120, 112, 95, 80, 105, 110, 115, 73, 110, 120, 70	Mean: 95.52, Median: 95.0, Mode: 120
As the manager, do you think which group perform better? Why?		

Questions to Consider: Additional Information



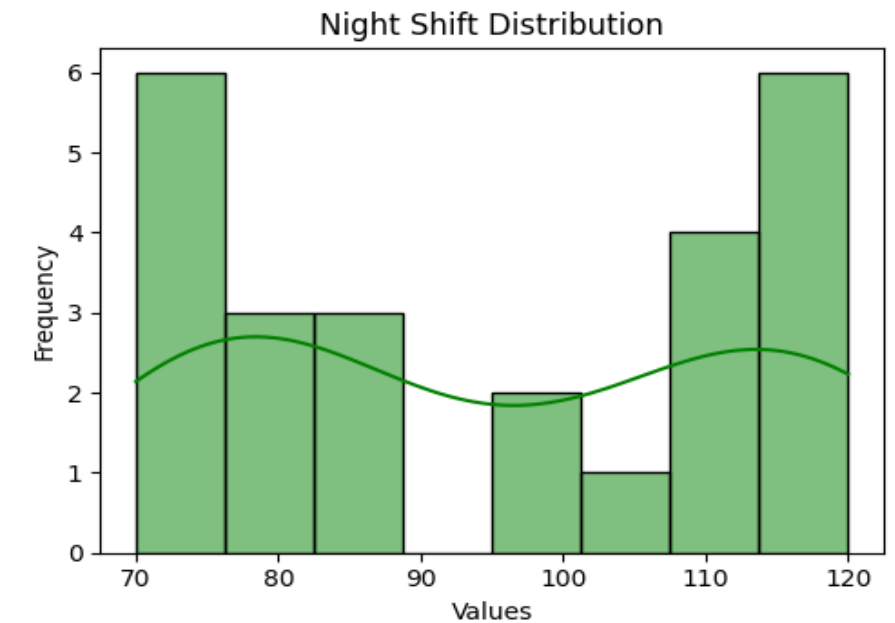
Morning Shift Statistics:

- Mean: 84.68
- Median: 85.0
- Mode: 87
- SD: 2.62
- Min: 80
- Max: 89



Afternoon Shift Statistics:

- Mean: 91.6
- Median: 90.0
- Mode: 90
- SD: 12.28
- Min: 65
- Max: 122



Night Shift Statistics:

- Mean: 95.52
- Median: 95.0
- Mode: 120
- SD: 18.52
- Min: 70
- Max: 120

With these additional information, would you still make the same decision? Still, is this ENOUGH to make an informed decision?

Excel Practices

Topics in This Session:

- Exploring data and visualizing distributions with histograms and fitting curves
- Calculating confidence intervals for means
- Performing One-sample t-Test using Excel
- Interpreting One-sample t-Test using (1) CI-Method and (2) p-value Method [Additional Method: Critical Value Method]

Business Scenario

A company has been tracking customer satisfaction scores for its range of products over the past few years. As the management team prepares for strategic planning, they want to evaluate whether customer satisfaction has remained consistent this year compared to previous years. To make informed decisions about marketing strategies, the team aims to analyze the current year's satisfaction scores against the historical data and other variables. **Guiding Questions:**

- What is the average customer satisfaction for this year compared to the previous years?
- Is there a difference between this year's satisfaction and the historical average? If so, what direction (higher or lower)?
- How do this year's satisfaction compared to previous year based on different factors, for example product categories (e.g., Product Return, Loyalty Member) and demographic factors (e.g., Gender, Income, ...)?

Additional Information:

- Overall Satisfaction =8.5
- By Gender: (M=8.8, F= 8.2, Other=8.4)
- By Loyalty Member: (Yes=8.8, No= 7.8)
- By Income: (Low=8.8, Medium= 8.2, High=8.1)
- By Product Return: (Less than 5%= 8.8, 5%-10%=8.5, More than 10%= 7.8)

The information was made up for study purpose only.

One Sample t-Test & Important Assumptions

A one-sample t-test is used when you want to compare the mean of a single sample to a known or hypothesized population mean. Here are the key situations and conditions for using a one-sample t-test:

Main Assumptions:

- **Independence of Observations:** the observations must be independent of each other. Each data point should not influence another.
- **Normality:** the data should be approximately normally distributed. This is particularly important when the sample size is small (typically $n < 30$).
- **Scale of Measurement:** The variable being tested should be measured on a continuous scale (interval or ratio).
- **Random Sampling:** the data should be collected through a process of random sampling to ensure that the sample is representative of the population.

Skewness & Kurtosis,
Q-Q plots, histograms
or Shapiro-Wilk test

Note: If any of these assumptions are violated, the results of the one-sample t-test may not be valid.

Understanding Data

Qualitative Variables (IV)		Test Variables (DV)
Variable Name	#Groups/Categories	
Gender	Three: M, F, & Other	Purchase_Amount, Product_Rating, Customer_Satisfaction Time_Decision Time_on Product
Income	Three: Low, Medium & High	
Marital_Status	Four: Single, Married, Widowed, Divorced	
Return_Rae	Three: Less than 5%, 5%-10% & More than 10%	
Customer_Loyalty	Two: Yes & No	

Data Analysis Plans		
Goal	Involved Variables	
	QUAN	QUAL
Descriptive Exploration of IV & DV	Mean, Mode, Median, SD, Boxplot	Bar, Pie, Frequency Table
	Boxplot by Factors	
Checking Assumptions DV	Skewness, Kurtosis, Histogram, Q-Q Plot, Shapiro-Test ...	
Inferential Statistics (Hypothesis Testing)	One-sample t-Tests, One-sample t-Tests by Variables	

One Sample t-Test: Decision Criteria

Positions of Test Value & p-value:

A. CI Method

- **Within CI:** No significant difference; fail to reject the null hypothesis. you do not have enough evidence to reject the null hypothesis (that the sample mean is equal to the population mean). This suggests that the sample mean is not significantly different from the population mean.
- **Outside CI:** Significant difference; reject the null hypothesis. you have enough evidence to reject the null hypothesis. This suggests that the sample mean is significantly different from the population mean.

B. p-value Method (Assume we take 95% CI)

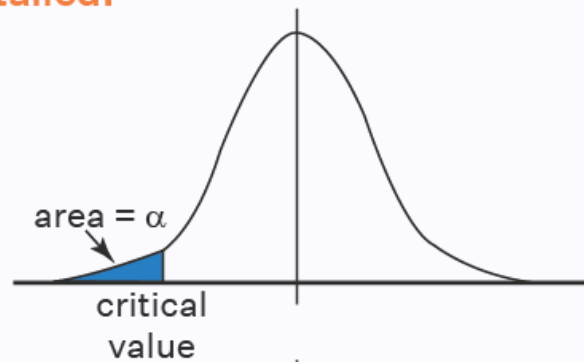
- If the p-value is greater than significance level ($\alpha = 0.05$), it indicates no significant difference (Same conclusion as Inside CI).
- If the p-value is less than 0.05 ($\alpha = 0.05$), it indicates a significant difference. (Same conclusion as Outside CI).

One Sample t-Test: Decision Criteria

Rejection Region for Null Hypothesis

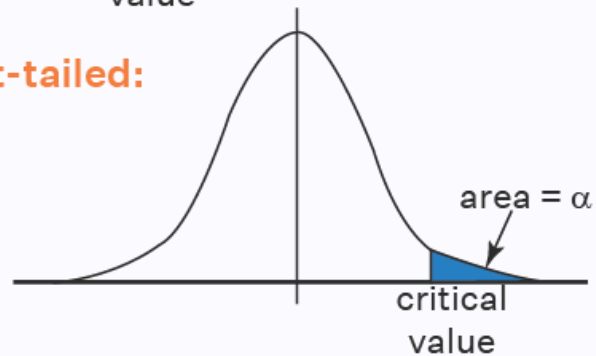


left-tailed:

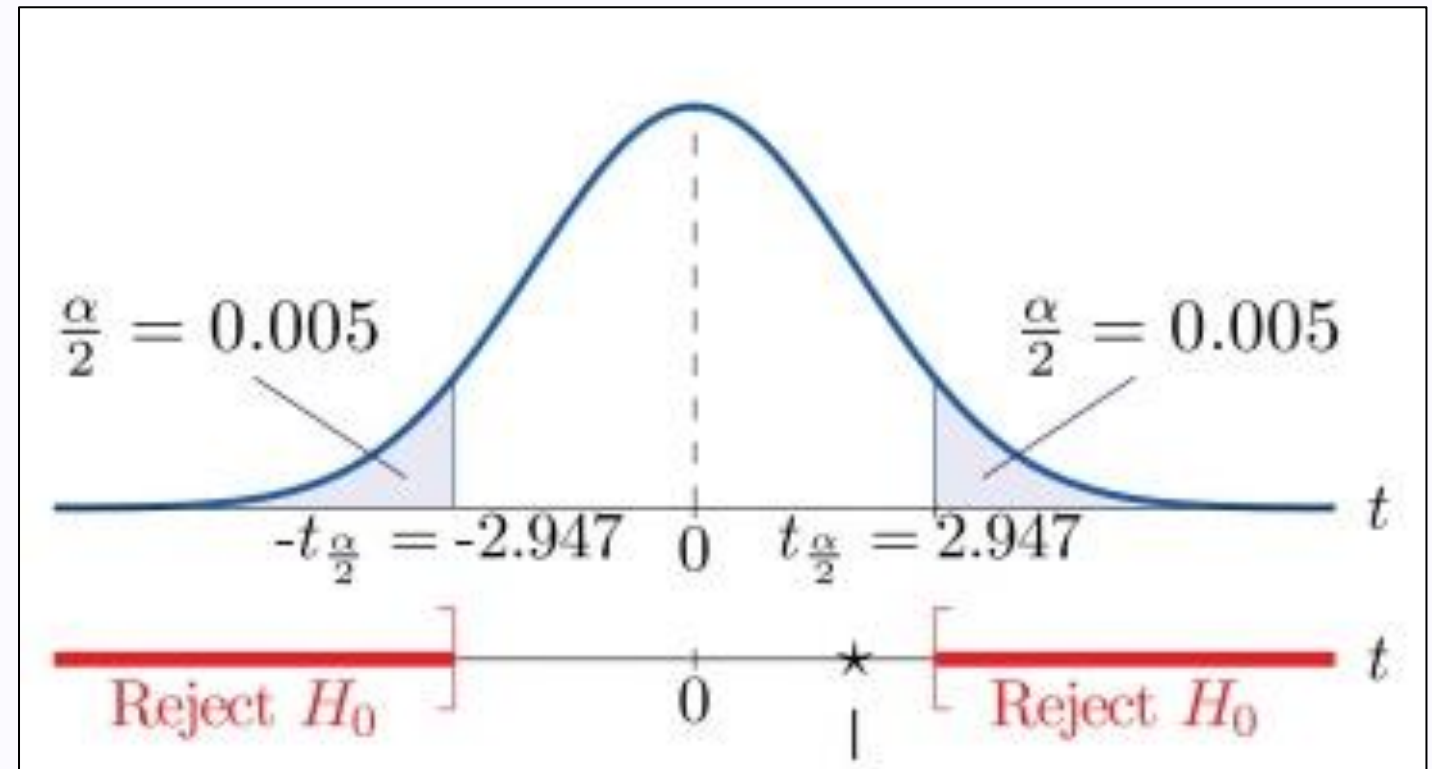
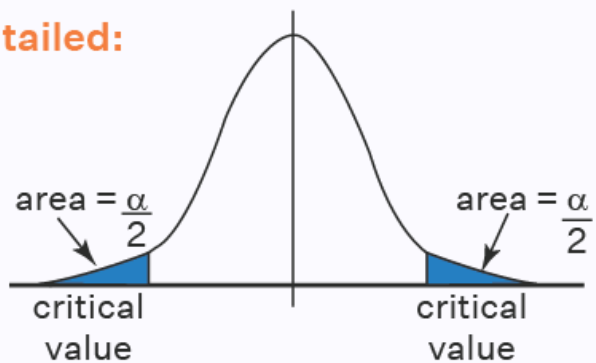


- - Reject H_0
- - Do not reject H_0

right-tailed:



two-tailed:



Interpreting the Results of One-sample t-Test

Check the p-value

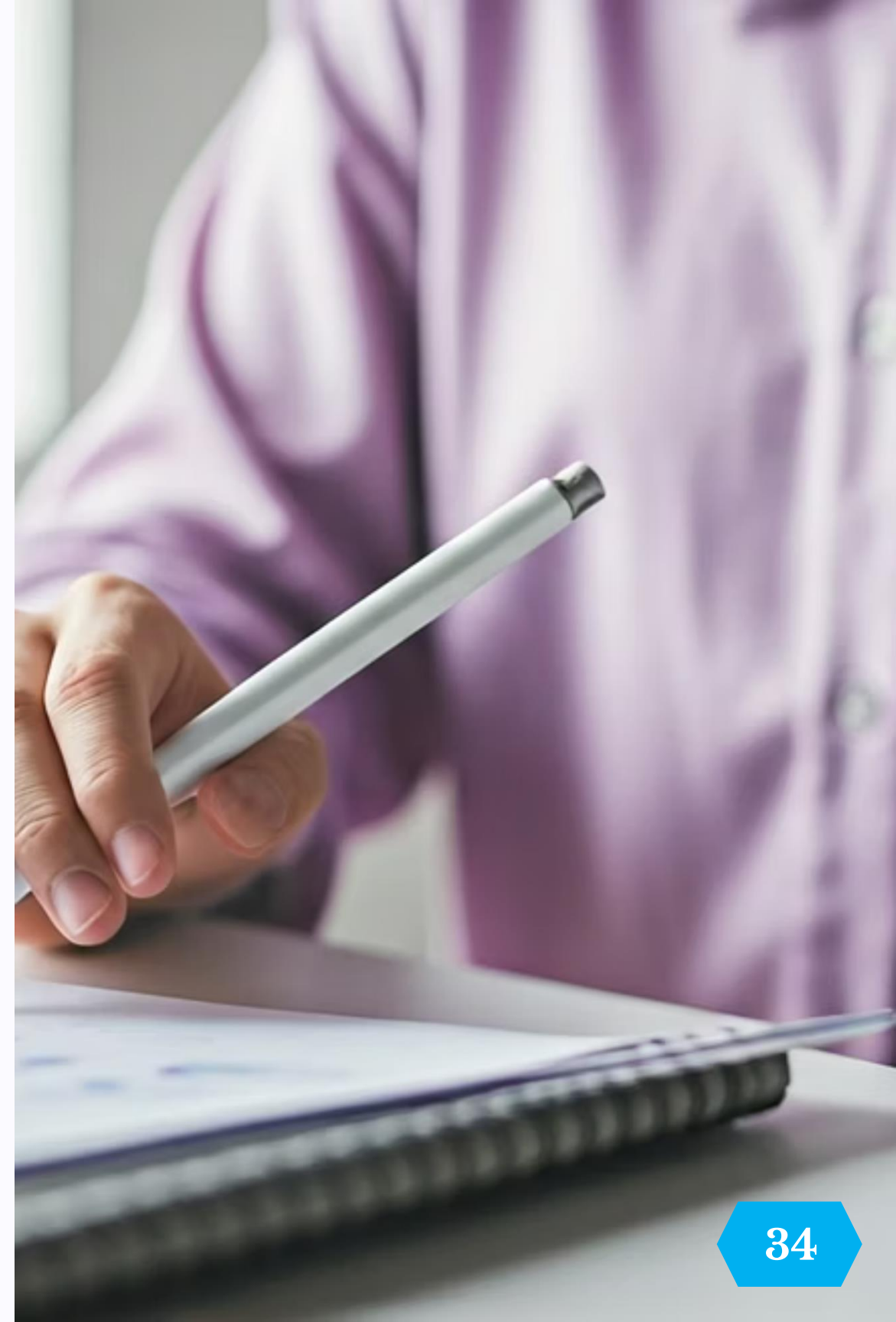
If $p < 0.05$ (or your chosen significance level), reject the null hypothesis that the sample mean equals the test value

Review Confidence Intervals

Check if the test value falls within the confidence interval of the sample mean

Check Effect Size

Check if the test value falls within the confidence interval of the sample mean



Interpreting the Result of One-sample t-Test

Results

T-Test

One-Sample Statistics				
	N	Mean	Std. Deviation	Std. Error Mean
Overall, how satisfied are you with your purchases?	1000	9.771	.6929	.0219

One-Sample Test						
Test Value = 8.5						
	t	df	Sig. (2-tailed)	Mean Difference	95% Confidence Interval of the Difference	
					Lower	Upper
Overall, how satisfied are you with your purchases?	57.987	999	.000	1.2705	1.228	1.313

Descriptives				
Overall, how satisfied are you with your purchases?			Statistic	Std. Error
	Mean		9.771	.0219
	95% Confidence Interval for Mean		Lower Bound	9.728
			Upper Bound	9.813

Effect Size

$$d = \frac{\bar{X} - \mu}{s} = \frac{9.77 - 8.5}{0.7} = 1.81$$

Interpretation

- *CI-Method*: the test value, 8.5 is Outside of CI=[9.813, 9.728] → Significant difference
→ Reject H_0 .

Conclusion: this year satisfaction is higher ($\bar{X} = 9.771 < 8.5$).

- *p-value Method*: from table, $p = 0.000 < 0.05$, provide evidence of statistically significant → Same conclusion as above.
- Effect Size $d = 1.81 > 1$ → large difference
→ large practical significance.

Interpretation of Effect Size

Effect Size Category	Cohen's d Range	Interpretation
Negligible	$-0.2 < d < 0.2$	Very small or trivial difference
Small	$-0.5 < d \leq -0.2$ or $0.2 \leq d < 0.5$	Small but noticeable difference
Medium	$-0.8 < d \leq -0.5$ or $0.5 \leq d < 0.8$	Moderate difference
Large	$d \leq -0.8$ or $d \geq 0.8$	Substantial difference

Interpretation of Cohen's d Greater than 1

- Cohen's $d > 1$:
 - Indicates a very large effect size, meaning the difference between the sample mean and the population mean is more than one standard deviation. This suggests a substantial and possibly highly significant difference.